

User Acceptance Testing (UAT) Checklist

Rwanda NBSAP Monitoring Dashboard 2025–2030

Prepared for: Rwanda Environment Management Authority (REMA) **Document Reference:** NBSAP-UAT-2026-010 **Version:** 1.0 **Date:** June 2026 **Classification:** Official — Government Use

Document Control

Field	Value
Document Title	UAT Checklist
Project	Rwanda NBSAP Monitoring Dashboard
System URL	https://nbsap-dashboard-rw.vercel.app
Prepared by	NBSAP Dashboard Development Team
Approved by	REMA — Department of Biodiversity & Landscape Management
Status	Final

1. Testing Instructions

1.1 Purpose

This checklist is designed for government acceptance testers to systematically verify that the NBSAP Monitoring Dashboard meets all functional and non-functional requirements before formal sign-off.

1.2 How to Use This Checklist

- Test each item using the live system at <https://nbsap-dashboard-rw.vercel.app>
- Mark each item as PASS, FAIL, or N/A
- For FAIL items, record the observed behaviour and expected behaviour in the Notes column
- All FAIL items must be resolved before final acceptance

1.3 Test Accounts Required

Role	Purpose
REMA Administrator	Test full admin capabilities
Lead Government Ministry Reporter	Test reporting and verification
District Reporting Officer	Test district-level data entry
Policy Monitor	Test read-only access
Development Partner	Test analytical access
Public Viewer	Test restricted public access

2. Authentication & Account Management

#	Test Case	Expected Result	PASS/FAIL	Notes
AUTH-01	Navigate to login page	Login form displayed with email and password fields		
AUTH-02	Login with valid credentials	Redirected to dashboard; user name displayed in header		
AUTH-03	Login with invalid password	Error message displayed; no access granted		
AUTH-04	Login with non-existent email	Error message displayed; no access granted		
AUTH-05	Login with suspended account	Access denied with suspension message displayed		
AUTH-06	Login with deactivated account	Access denied with deactivation message displayed		
AUTH-07	Session persistence	Refresh page; user remains logged in		
AUTH-08	Logout	User returned to login page; session cleared		
AUTH-09	Access protected page without login	Redirected to login page		

3. Role-Based Access Control

3.1 REMA Administrator Access

#	Test Case	Expected Result	PASS/FAIL	Notes
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RBAC-01	Access Dashboard	Full dashboard displayed with all metrics		
RBAC-02	Access User Management	User list displayed with edit/suspend/deactivate controls		
RBAC-03	Access Verification Queue	All pending submissions visible with approve/reject controls		
RBAC-04	Access Audit Log	Complete audit trail visible with filters		
RBAC-05	Access Reporting Toolkit	All 7 tools (T01–T07) available for submission		
RBAC-06	Access Role Requests	Role change request queue visible		
RBAC-07	Export data (CSV/PDF/JSON)	Export completes successfully; file downloads		

3.2 District Reporting Officer Access

#	Test Case	Expected Result	PASS/FAIL	Notes
RBAC-08	Access Dashboard	Dashboard displayed (read-only metrics)		
RBAC-09	Access Reporting Toolkit	Tools available for data submission		
RBAC-10	Attempt to access User Management	Access denied; unauthorised page displayed		
RBAC-11	Attempt to access Audit Log	Access denied; unauthorised page displayed		
RBAC-12	Attempt to access Verification Queue	Access denied; unauthorised page displayed		

3.3 Public Viewer Access

#	Test Case	Expected Result	PASS/FAIL	Notes
RBAC-13	Access Dashboard	Dashboard displayed (view-only)		
RBAC-14	Access National Targets	Targets displayed (view-only)		
RBAC-15	Access Map	Interactive map displayed		

RBAC-16	Attempt to access Reporting Toolkit	Access denied		
RBAC-17	Attempt to access Verification Queue	Access denied		
RBAC-18	Attempt to access User Management	Access denied		
RBAC-19	Attempt to export raw data	Export option not available		

4. Dashboard

#	Test Case	Expected Result	PASS/FAIL	Notes
DSH-01	Dashboard loads	All metric cards displayed with values		
DSH-02	13 automated metrics visible	All metric cards show non-null values		
DSH-03	Progress colour coding correct	Green ($\geq 70\%$), Amber (40-69%), Red ($< 40\%$)		
DSH-04	NBSAP Target progress section	22 targets listed with progress bars		
DSH-05	Target drill-down	Clicking a target shows details and linked indicators		
DSH-06	Recent approved reports	Latest approved submissions displayed		
DSH-07	Real-time updates	Approving a report updates dashboard metrics without refresh		
DSH-08	Page loads within 3 seconds	Measure using browser DevTools Network tab		

5. National Targets

#	Test Case	Expected Result	PASS/FAIL	Notes
TGT-01	All 22 targets displayed	Targets 1–22 listed, organised by Goal A/B/C/D		
TGT-02	Goal A targets (1–8)	Correctly grouped under "Reduce Threats to Biodiversity"		

TGT-03	Goal B targets (9–12)	Correctly grouped under "Meet People's Needs"		
TGT-04	Goal C targets (13–17)	Correctly grouped under "Tools and Solutions"		
TGT-05	Goal D targets (18–22)	Correctly grouped under "Means of Implementation"		
TGT-06	Target details	Each target shows title, description, baseline, progress		
TGT-07	Linked indicators	Each target displays its associated indicators		
TGT-08	Implementation milestones	2025–2030 milestones displayed per target		
TGT-09	Responsible stakeholders	Stakeholder list shown for each target		
TGT-10	Progress bars accurate	Progress matches approved report data		

6. Indicators

#	Test Case	Expected Result	PASS/FAIL	Notes
IND-01	All ~79 indicators displayed	Complete indicator list with tier classification		
IND-02	Tier classification correct	Headline, Component, Binary, Complementary tiers shown		
IND-03	Indicator details	Each shows: name, definition, baseline, current value, 2030 target		
IND-04	Status badges	On-track (green), At-risk (amber), Behind (red) correctly colour-coded		
IND-05	Parent target linkage	Each indicator shows its parent NBSAP target		
IND-06	Search functionality	Searching by indicator name/number filters the list correctly		
IND-07	Filter by Goal	Filtering by Goal A/B/C/D shows only relevant indicators		

7. Reporting Toolkit (T01–T07)

7.1 Report Submission

#	Test Case	Expected Result	PASS/FAIL	Notes
RPT-01	T01 form loads	National Institutional Reporting form displayed with all fields		
RPT-02	T02 form loads	Ecosystem & Habitat Monitoring form displayed		
RPT-03	T03 form loads	Protected Area Management form displayed		
RPT-04	T04 form loads	Human-Wildlife Conflict form displayed		
RPT-05	T05 form loads	Finance & Resource Mobilisation form displayed		
RPT-06	T06 form loads	Private Sector EIA Compliance form displayed		
RPT-07	T07 form loads	Community Engagement form displayed		
RPT-08	Submit with all required fields	Submission accepted; status = 'pending'		
RPT-09	Submit with missing required fields	Validation error displayed; submission blocked		
RPT-10	Attach evidence file	File uploads successfully; appears in attachments		
RPT-11	Select district	District dropdown shows all 30 districts		
RPT-12	Select NBSAP target	Target dropdown shows all 22 targets		
RPT-13	View submission history	User's previous submissions displayed with status		

7.2 Data Validation

#	Test Case	Expected Result	PASS/FAIL	Notes
VAL-01	Numeric field with text input	Validation error; field rejects non-numeric input		
VAL-02	Percentage field > 100	Validation error; value must be 0–100		

VAL-03	Negative hectare value	Validation error; hectares must be ≥ 0		
VAL-04	Empty required field	Validation error; field highlighted as required		
VAL-05	Date field with invalid date	Validation error or calendar picker prevents invalid dates		

8. Verification Queue

#	Test Case	Expected Result	PASS/FAIL	Notes
VRF-01	Queue displays pending reports	All pending submissions listed with submitter, tool, date		
VRF-02	View submission details	Full form data and evidence attachments visible		
VRF-03	Approve a report	Status changes to 'approved'; reviewer ID and timestamp recorded		
VRF-04	Reject a report	Rejection reason required; status changes to 'rejected'		
VRF-05	Reject without reason	System prevents rejection; reason field required		
VRF-06	Metrics update on approval	System metrics update immediately after approval		
VRF-07	Notification to reporter on approval	Reporter receives in-app notification		
VRF-08	Notification to reporter on rejection	Reporter receives notification with rejection reason		
VRF-09	Approved report shows reviewer info	Reviewer name and timestamp displayed on report		
VRF-10	Delete approved report	Metrics auto-reversed; audit log entry created		

9. Interactive Map

#	Test Case	Expected Result	PASS/FAIL	Notes
MAP-01	Map renders	Interactive map of Rwanda displayed		

MAP-02	All 30 districts visible	District boundaries rendered correctly		
MAP-03	Protected areas layer	Toggle on: protected area polygons displayed		
MAP-04	River network layer	Toggle on: river network lines displayed		
MAP-05	Lakes layer	Toggle on: lake boundaries displayed		
MAP-06	GBIF species occurrences	Species occurrence points displayed on map		
MAP-07	Layer toggle on/off	Each layer can be independently toggled		
MAP-08	District click/hover	District detail panel shows compliance and forest cover		
MAP-09	Map zoom and pan	Smooth zoom and pan interaction		
MAP-10	Map legend	Legend displays colour coding for active layers		
MAP-11	Map export	Map can be exported as image		

10. RBIS Dashboard

#	Test Case	Expected Result	PASS/FAIL	Notes
RBIS-01	RBIS page loads	Connection bar, metrics, indicators matrix displayed		
RBIS-02	Connection status	Shows current RBIS connection status (connected/disconnected)		
RBIS-03	Metrics panel	Total occurrences, 24h/7d counts, active streams displayed		
RBIS-04	Indicators matrix	All 79 indicators listed by Goal/Target with progress bars		
RBIS-05	Search indicators	Search filters indicators by name/description/number		
RBIS-06	Filter by Goal	Clicking Goal A/B/C/D filters the matrix		

RBIS-07	8 data streams displayed	All data streams shown with occurrence counts and status		
RBIS-08	Target badge click	Clicking target badge in Signal Feed scrolls to that target		
RBIS-09	RBIS linkage badges	Indicators show linked/not-linked/partial status		
RBIS-10	Auto-refresh	Metrics refresh every 30s; data streams every 60s		

11. Data Export

#	Test Case	Expected Result	PASS/FAIL	Notes
EXP-01	Export CSV	CSV file downloads with correct data and headers		
EXP-02	Export PDF	Formatted PDF report generates and downloads		
EXP-03	Export JSON	Structured JSON file downloads with correct schema		
EXP-04	Export filtered data	Export respects active filters (date range, tool, district)		
EXP-05	Export audit log entry	Export action logged in audit trail		
EXP-06	Non-authorized user export	Export option not available for restricted roles		

12. User Management (Admin Only)

#	Test Case	Expected Result	PASS/FAIL	Notes
USR-01	View all users	Complete user list with roles, status, last login		
USR-02	Change user role	Role updated; user's access changes immediately		
USR-03	Suspend user with end date	Account suspended; shows suspension reason and end date		
USR-04	Auto-reactivation after end date	Suspended account reactivated when end date passes		

USR-05	Deactivate user	Account permanently deactivated; login blocked		
USR-06	Role change request approval	Request approved; user role updated; notification sent		
USR-07	Role change request rejection	Request rejected with reason; notification sent		
USR-08	Audit log entry for user actions	All user management actions logged in audit trail		

13. Audit Log

#	Test Case	Expected Result	PASS/FAIL	Notes
AUD-01	Audit log page loads (Admin only)	Complete log displayed with filters		
AUD-02	Report submission logged	New entry with action_type = 'submit'		
AUD-03	Report approval logged	New entry with action_type = 'approve'		
AUD-04	Report rejection logged	New entry with action_type = 'reject'		
AUD-05	User login logged	New entry with action_type = 'login'		
AUD-06	Data export logged	New entry with action_type = 'export'		
AUD-07	Log immutability	Audit log entries cannot be edited or deleted		
AUD-08	Log contains required fields	Each entry: user ID, action type, action, timestamp, role		

14. Notifications

#	Test Case	Expected Result	PASS/FAIL	Notes
NTF-01	Notification on report approval	Reporter receives in-app notification		
NTF-02	Notification on report rejection	Reporter receives notification with reason		

NTF-03	Notification on role change	User notified of role change approval/rejection		
NTF-04	Unread badge	Notification bell shows unread count		
NTF-05	Mark as read	Clicking notification marks it as read		
NTF-06	Real-time delivery	Notification appears without page refresh (WebSocket)		

15. Non-Functional Requirements

15.1 Performance

#	Test Case	Expected Result	PASS/FAIL	Notes
PERF-01	Initial page load	< 3 seconds on 4G connection		
PERF-02	Dashboard metric load	Metrics display within 2 seconds		
PERF-03	Map render time	Map renders within 2 seconds		
PERF-04	Search response	Search results appear within 500ms		
PERF-05	Report submission	Form submits within 2 seconds		

15.2 Security

#	Test Case	Expected Result	PASS/FAIL	Notes
SEC-01	HTTPS enforced	All pages served over HTTPS; HTTP redirects to HTTPS		
SEC-02	Security headers present	CSP, X-Frame-Options, X-Content-Type-Options in response headers		
SEC-03	Direct API access blocked	Attempting direct Supabase API call without auth returns 401		
SEC-04	RLS enforcement	User cannot access another user's data via API manipulation		
SEC-05	Password not stored in plaintext	Passwords hashed (bcrypt) in database		

15.3 Responsiveness

#	Test Case	Expected Result	PASS/FAIL	Notes
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RES-01	Desktop (1920x1080)	Full layout displayed correctly		
RES-02	Tablet (768x1024)	Responsive layout; all features accessible		
RES-03	Mobile (375x667)	Mobile layout; navigation accessible via menu		
RES-04	Map on mobile	Map renders and is interactive on mobile devices		

15.4 Browser Compatibility

#	Test Case	Expected Result	PASS/FAIL	Notes
BRW-01	Chrome 90+	All features work correctly		
BRW-02	Firefox 88+	All features work correctly		
BRW-03	Safari 14+	All features work correctly		
BRW-04	Edge 90+	All features work correctly		
BRW-05	Chrome Mobile	All features work correctly on mobile		

16. Sign-Off

16.1 Test Summary

Category	Total Tests	Pass	Fail	N/A
Authentication	9			
RBAC	19			
Dashboard	8			
National Targets	10			
Indicators	7			
Reporting Toolkit	18			
Verification Queue	10			
Map	11			
RBIS Dashboard	10			
Data Export	6			
User Management	8			
Audit Log	8			
Notifications	6			

Non-Functional	14			
TOTAL	144			

16.2 Acceptance Decision

Decision	Signature	Date
Accepted / Accepted with conditions / Not accepted		

Tester Name: _____

Tester Role: _____

Organisation: _____

Comments:

16.3 Outstanding Issues (if any)

#	Issue	Severity	Resolution Required By
1			
2			
3			

*Prepared by: NBSAP Dashboard Development Team Document Reference: NBSAP-UAT-2026-010
Rwanda Environment Management Authority (REMA)*